



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

November 2007

First Assessment

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question

Total Marks – 60

- All Questions may be attempted
- Each Question is worth 15 marks, and should be handed up in a separate examination Booklet.
- Full Marks may not be awarded for careless or poorly set out work.

Examiner – *A.M. Gainford*

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Question 1. (15 Marks) (Start a new booklet.)

- | | Marks |
|--|--------------|
| (a) (i) In how many ways may all the letters of the word ANAGRAM be arranged in a line? | 2 |
| (ii) In how many ways may four men and four women be arranged around a circular table, if no two men are to be seated next to each other? | |
| (b) The point $P(8, -16)$ divides the interval AB externally in the ratio 2:1. Find the coordinates of the point B given that A is $(-2, 6)$. | 3 |
| (c) Solve the following inequalities, and in each case graph the solution on the number line: | 6 |
| (i) $\frac{2}{x} > \frac{1}{3}$ | |
| (ii) $\frac{x}{x-1} \leq 2$ | |
| (d) Consider the equation $3\sin \theta + 4\cos \theta = 2$. | 4 |
| (i) Express the equation in the form $R\sin(\theta + \alpha) = 2$, where R is a constant, and α is constant and acute. | |
| (ii) Hence or otherwise solve the equation for $0^\circ \leq \theta \leq 360^\circ$, correct to the nearest minute. | |

Question 2. (15 Marks) (Start a new booklet.)

Marks

- (a) Solve the following equation for $0^\circ \leq \theta \leq 360^\circ$:

2

$$6 \cos^2 \theta = 4 + \sin \theta.$$

Give your answers correct to the nearest minute.

- (b) Given the polynomial $P(x) = 2x^3 - 6x^2 - 12x + 16$:

4

(i) Use the factor theorem to determine a zero of the polynomial.

(ii) Express $P(x)$ as a product of linear factors.

- (c) Show that:

2

$$2 \sec \theta (\sec \theta - \tan \theta) - 1 = \frac{1 - \sin \theta}{1 + \sin \theta}$$

- (d) Find the acute angle between the lines $x + 2y - 6 = 0$ and $3x - 2y + 7 = 0$, correct to the nearest minute.

3

- (e) Write in polynomial form the equation of the monic quartic with a double root at $x = 2$, and simple roots at $x = 0$ and $x = -3$.

2

- (f) When a certain polynomial is divided by $(x + 2)(x - 3)$ the remainder is $2x + 3$. What are the remainders when it is divided by $x + 2$ and $x - 3$ separately?

2

Question 3. (15 Marks) (Start a new booklet.)

Marks

- (a) Solve the equation $x^4 + 4x^3 - 16x - 16 = 0$ by factorization, or otherwise. **3**
- (b) From a point A on the ground due South of the base G of a television tower, the angle of elevation of the top (T) of the tower is $16^\circ 42'$. One kilometre East of A , at the point B , the angle of elevation of the top of the tower is 12° . **4**
- (i) Draw a neat diagram representing this situation.
- (ii) Find the height of the tower to the nearest metre.
- (c) P is a point on the parabola $x^2 = 4ay$ and G is the point where the normal at P intersects the axis of the parabola. If the line through P perpendicular to the axis of the parabola meets the axis at N , then the interval NG is called the subnormal corresponding to P . **4**
- (i) Show that the point $P(2ap, ap^2)$ lies on the parabola for all values of p .
- (ii) Derive the equation of the normal at P .
- (iii) Prove that the length of the subnormal corresponding to P is constant, and equal to twice the focal length.
- (d) Solve the inequality $|2x - 1| > |x + 2|$. **2**
- (e) Find the general solution of the equation $\cos \theta \sin \theta = \frac{1}{2}$. **2**

Question 4. (15 Marks) (Start a new booklet.)

Marks

- (a) Five persons are to be seated in a row. **3**
- (i) How many arrangements are possible?
- (ii) What is the probability that person A and person B are separated by exactly one other person?
- (b) P is a variable point on the parabola $x^2 = 4ay$. The tangent at P meets the x -axis at Q and the y -axis at R . **4**
Determine the Cartesian equation of the locus of M , the midpoint of QR as P moves on the parabola.
- (c) The ropes of a swing are 4m long and at rest the seat is 0.6 m above the ground. A child uses the swing, and the highest point reached on one side is 2m, and on the other side is 2.3 m above the ground. **4**
- (i) Calculate, to the nearest degree, the angle through which the seat swings.
- (ii) Find, to the nearest centimetre the straight line distance between the two highest points.
- (d) (i) Find an expression for $\tan(\alpha + \beta + \gamma)$ in terms of $\tan \alpha$, $\tan \beta$, $\tan \gamma$. **4**
- (ii) Put $\alpha = \beta = \gamma = \frac{\pi}{12}$ in the expression derived above, and show that $\tan \frac{\pi}{12}$ is a root of the equation $t^3 - 3t^2 - 3t + 1 = 0$.

This is the end of the paper.

2007 Mathematics Extension 1 Assessment 1: Solutions Question 1

1. (a) (i) In how many ways can the letters of the word ANAGRAM be arranged in a line? 1

Solution: $\frac{7!}{3!} = \frac{5040}{6},$
 $= 840.$

- (ii) In how many ways can four men and four women be arranged around a circular table, if no two men are to be seated next to each other? 1

Solution: $3!4! = 144.$

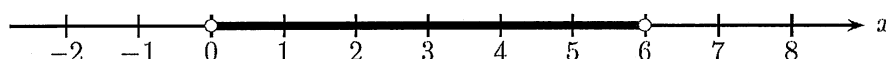
- (b) The point $P(8, -16)$ divides the interval AB externally in the ratio $2 : 1$. Find the coordinates of the point B given that A is $(-2, 6)$. 3

Solution: $8 = \frac{(-1) \times (-2) + 2 \times x}{-1 + 2}, \quad -16 = \frac{(-1) \times 6 + 2 \times y}{-1 + 2},$
 $= 2 + 2x, \quad = -6 + 2y,$
 $\therefore x = 3. \quad y = -5.$
i.e. $B(3, -5).$

- (c) Solve the following inequalities, and in each case graph the solution on the number line:

(i) $\frac{2}{x} > \frac{1}{3}$ 3

Solution: Clearly $x > 0$ (as $1/3 > 0$), $\therefore 6 > x,$
i.e. $0 < x < 6.$



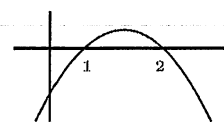
(ii) $\frac{x}{x-1} \leq 2$ 3

Solution: If $x - 1 > 0$, $x > 1$; if $x - 1 < 0$, $x < 1$;
then $x \leq 2x - 2$, then $x \geq 2x - 2$,
 $x \geq 2.$ $x \leq 2.$
 $\therefore x \geq 2.$ $\therefore x < 1.$



Alternative Solution: $x(x-1) \leq 2(x-1)^2,$
 $(x-1)(x-2x+2) \leq 0,$
 $(x-1)(2-x) \leq 0.$
 $\therefore x < 1$ or $x \geq 2.$

Number line graph as above.



(d) Consider the equation $3 \sin \theta + 4 \cos \theta = 2$.

- (i) Express the equation in the form $R \sin(\theta + \alpha) = 2$, where R is a constant, and α is constant and acute. 2

Solution: $R = \sqrt{3^2 + 4^2}, \quad \tan \alpha = \frac{4}{3},$
 $\quad \quad \quad = 5. \quad \quad \quad \alpha \approx 53.130102354^\circ.$
i.e. $5 \sin(\theta + \tan^{-1}(\frac{4}{3})) = 2.$

- (ii) Hence or otherwise solve the equation for $0^\circ \leq \theta \leq 360^\circ$, correct to the nearest minute. 2

Solution: $\sin(\theta + 53^\circ 7' 48.3685'') \approx \frac{2}{5},$
 $\theta + 53^\circ 7' 48.3685'' \approx 23^\circ 34' 41.4425'', 156^\circ 25' 18.5575'',$
 $\quad \quad \quad 383^\circ 34' 41.4425'', \dots$
 $\theta \approx 103^\circ 17' 30.1890'', 330^\circ 26' 53.0740'',$
 $\quad \quad \approx 103^\circ 18', 330^\circ 27' \text{ (nearest minute).}$

2) a)

$$6\cos^2\theta = 4 + \sin\theta$$

$$6(1 - \sin^2\theta) = 4 + \sin\theta$$

$$6 - 6\sin^2\theta = 4 + \sin\theta$$

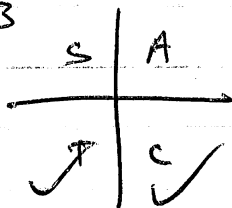
$$6\sin^2\theta + \sin\theta - 2 = 0$$

$$(6\sin\theta + 4)(6\sin\theta - 3) = 0$$

$$\begin{matrix} 6 \\ 2 \times 3 \end{matrix}$$

$$(3\sin\theta + 2)(2\sin\theta - 1) = 0$$

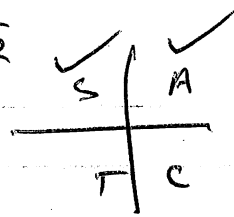
$$\sin\theta = -\frac{2}{3}$$



$$\sin\alpha = \frac{2}{3}$$

$$\alpha = 41^\circ 49'$$

$$\sin\theta = \frac{1}{2}$$



$$\sin\alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$

$$\theta = 30^\circ, 150^\circ, 221^\circ 49', 318^\circ 11'$$

b) i) $P(x) = 2x^3 - 6x^2 - 12x + 16$

$$P(1) = 2(1)^3 - 6(1)^2 - 12(1) + 16 = 0$$

$\therefore x=1$ is a zero

ii) $\therefore x-1$ is a factor

$$\begin{array}{r} 2x^2 - 4x - 16 \\ x-1 \overline{) 2x^3 - 6x^2 - 12x + 16} \\ \underline{2x^3 - 2x^2} \\ -4x^2 - 12x \\ \underline{-4x^2 + 4x} \\ -16x + 16 \\ \underline{-16x + 16} \\ 0 \end{array}$$

$$\therefore P(x) = (x-1)(2x^2 - 4x - 16)$$

$$P(x) = 2(x-1)(x^2 - 2x - 8)$$

$$\begin{matrix} x & -8 \\ + & \underline{-2} \\ & -4, 2 \end{matrix}$$

$$P(x) = 2(x-1)(x+2)(x-4)$$

$$\begin{aligned}
 c) \quad LHS &= 2\sec\theta(\sec\theta - \tan\theta) - 1 \\
 &= \frac{2}{\cos\theta} \left(\frac{1}{\cos\theta} - \frac{\sin\theta}{\cos\theta} \right) - 1 \\
 &= \frac{2(1 - \sin\theta)}{\cos^2\theta} - 1 \\
 &= \frac{2(1 - \sin\theta)}{1 - \sin^2\theta} - 1 \\
 &= \frac{2(1 - \sin\theta)}{(1 - \sin\theta)(1 + \sin\theta)} - 1 \\
 &= \frac{2}{1 + \sin\theta} - 1 \\
 &= \frac{2 - (1 + \sin\theta)}{1 + \sin\theta} \\
 &= \frac{1 - \sin\theta}{1 + \sin\theta} \\
 &= RHS
 \end{aligned}$$

$$\therefore 2\sec\theta(\sec\theta - \tan\theta) - 1 = \frac{1 - \sin\theta}{1 + \sin\theta}$$

$$\begin{aligned}
 d) \quad x + 2y - 6 &= 0 \\
 2y &= -x + 6 \\
 y &= -\frac{x}{2} + 3 \\
 m_1 &= -\frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 3x - 2y + 7 &= 0 \\
 2y &= 3x + 7 \\
 y &= \frac{3}{2}x + \frac{7}{2} \\
 m_2 &= \frac{3}{2}
 \end{aligned}$$

$$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{\left(-\frac{1}{2}\right) - \left(\frac{3}{2}\right)}{1 + \left(-\frac{1}{2}\right)\left(\frac{3}{2}\right)} \right|$$

$$\tan \theta = |-8|$$

$$\tan \theta = 8$$

$$\theta = 82^{\circ}52'$$

$$\begin{aligned} \text{e)} \quad (x-2)^2 x (x+3) &= 0 \\ (x^2 - 4x + 4)(x^2 + 3x) &= 0 \\ x^4 - 4x^3 + 4x^2 + 3x^3 - 12x^2 + 12x &= 0 \\ x^4 - x^3 - 8x^2 + 12x &= 0 \end{aligned}$$

$$\text{f)} \quad P(x) = (x+2)(x-3)Q(x) + 2x+3$$

$$\begin{aligned} P(-2) &= 2(-2) + 3 \\ &= -1 \end{aligned}$$

$$\begin{aligned} P(3) &= 2(3) + 3 \\ &= 9 \end{aligned}$$

$\therefore$ the remainders when $P(x)$ is divided by $x+2$ and $x-3$ are -1 and 9 respectively.

QUESTION 3. (EXT 1 VRII 2007)

(a) $x^4 + 4x^3 - 16x - 16 = 0.$

$\therefore x^4 - 16 + 4x^3 - 16x = 0$

$(x^2 - 4)(x^2 + 4) + 4x(x^2 - 4) = 0.$

$(x^2 - 4)(x^2 + 4x + 4) = 0$

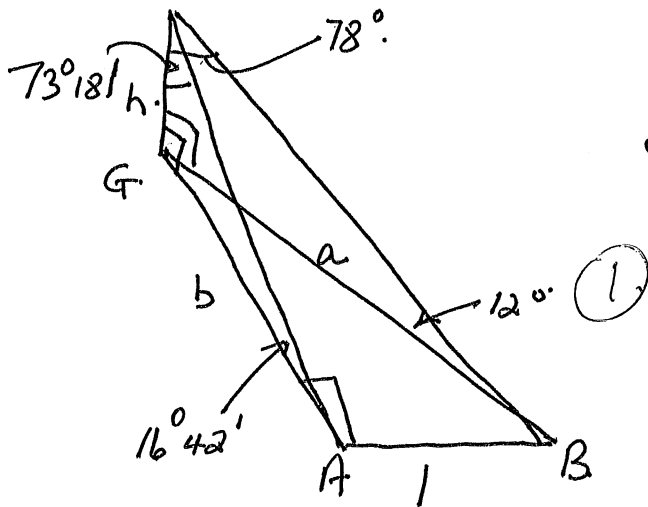
$(x - 2)(x + 2)(x + 2)^2 = 0$

$(x - 2)(x + 2)^3 = 0$

$\boxed{x = 2, -2, -2, -2.}$

(3)

(b)



$b = h \tan 73^\circ 18'$

$\& a = h \tan 78^\circ.$

$a^2 - b^2 = 1.$

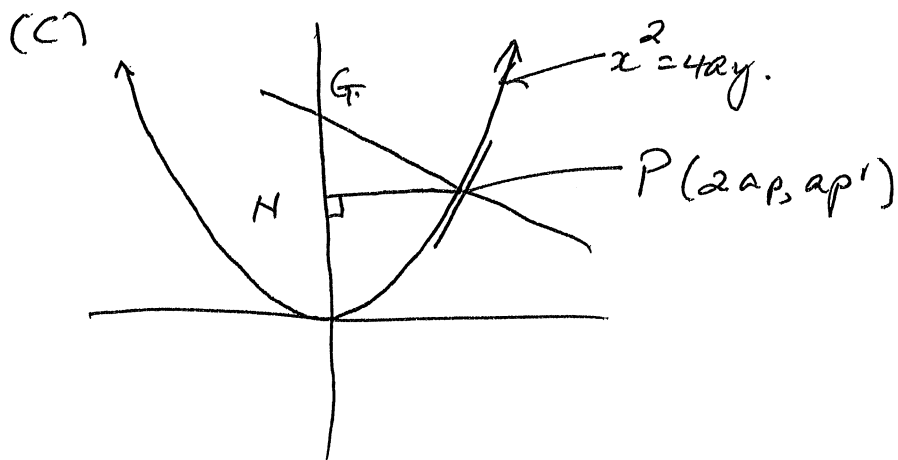
$\therefore h^2 [\tan^2 78^\circ - \tan^2 73^\circ 18'] = 1$

$h^2 = \frac{1}{\tan^2 78^\circ - \tan^2 73^\circ 18'}$

$h = \frac{1}{\sqrt{\tan^2 78^\circ - \tan^2 73^\circ 18'}}$

$\boxed{h \doteq 301 \text{ m.}}$

(4)



$$\begin{aligned}
 \text{(i)} \quad x^2 &= (2ap)^2 \\
 &= 4a^2 p^2 \\
 &= 4a(ap^2) \\
 \therefore x^2 &= 4ay.
 \end{aligned}$$

$\therefore (2ap, ap^2)$
lies on
 $x^2 = 4ay.$

①

$$\begin{aligned}
 \text{(ii)} \quad y &= \frac{1}{4a} x^2 \\
 y' &= \frac{1}{2a} x.
 \end{aligned}$$

$$\begin{aligned}
 \therefore m_T &= \frac{1}{2a} \cdot 2ap \\
 &= p.
 \end{aligned}$$

$$\therefore m_N = -\frac{1}{p}.$$

$$\therefore N: y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$\begin{aligned}
 py - ap^3 &= -x + 2ap \\
 \boxed{x + py} &= 2ap + ap^3
 \end{aligned}$$

②

(iii) N is the point $(0, ap^2)$

$\therefore G$ is the point $(0, 2a + ap^2)$

$$\therefore NG = 2a.$$

①

Hence the subnormal NG is independent of P and equal to twice the focal length.

$$(d) \quad |2x-1| > |x+2|$$

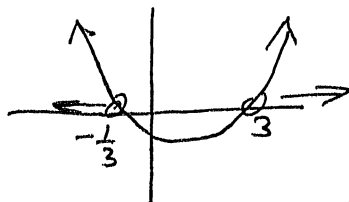
$$\therefore (2x-1)^2 > (x+2)^2$$

$$4x^2 - 4x + 1 > x^2 + 4x + 4$$

$$3x^2 - 8x - 3 > 0$$

$$(3x+1)(x-3) > 0$$

$$\therefore \boxed{x < -\frac{1}{3}, x > 3.}$$



(2)

$$(e) \quad \cos \theta \sin \theta = \frac{1}{2}$$

$$\therefore 2 \cos \theta \sin \theta = 1$$

$$\sin 2\theta = 1$$

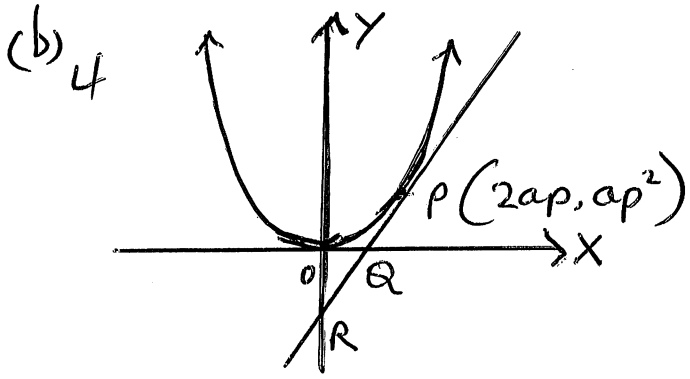
$$2\theta = n\pi + (-1)^n \frac{\pi}{2}$$

$$\boxed{\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}^*} \quad (2)$$

QUESTION 4

(a) (i) $5! = 120$ ways (1)

$$2 \text{ (ii)} \quad \frac{3 \times 2 \times 3!}{5!} = \frac{3}{10} \quad (2)$$



E_q^n tangent at P
 $y - px + ap^2 = 0$ (1)

When $x=0$, $y = -ap^2$
 $\Rightarrow R(0, -ap^2)$ 1/2

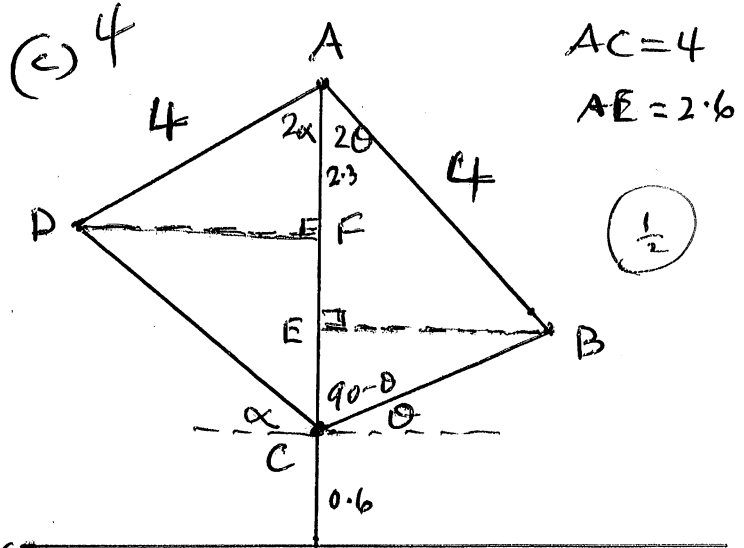
When $y=0$, $x=a_p$
 $\Rightarrow Q(a_p, 0)$ 1/2

$$M\left(\frac{ap}{2}, -\frac{ap^2}{2}\right) \left(\frac{1}{2}\right)$$

$$\text{let } x = \frac{ap}{2}, y = -\frac{ap^2}{2}$$

$$p = \frac{2x}{a}, \quad y = -\frac{a}{2} \left(\frac{2x}{a} \right)^2$$

locus $y = -\frac{2x^2}{a}$ $\left(\frac{1}{12}\right)$



(1)

In ΔADF $\cos 2\alpha = \frac{2.3}{4}$

$2\alpha = 54^\circ 54' 1.32''$

(1)

In ΔAEB $\cos 2\theta = \frac{2.6}{4}$

$2\theta = 49^\circ 27' 30.23''$

(1)

$$\angle DAB = 104^\circ$$

$$(BD)^2 = 4^2 + 4^2 - 2 \cdot 4 \cdot 4 \cos 104^\circ$$

$$BD = 6.3 \text{ m}$$

à 630 cm

(d)

$$(i) \tan(\alpha + \beta + \gamma) = \tan[(\alpha + \beta) + \gamma]$$

$$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} + \tan \gamma$$

$$1 - \left(\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \right) \cdot \tan \gamma$$

$$(*) \text{ or } \tan(\alpha + \beta + \gamma) = \frac{\tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma}{1 - \tan \alpha \tan \beta - \tan \beta \tan \gamma - \tan \alpha \tan \gamma}$$

$$(ii) \alpha = \beta = \gamma = \frac{\pi}{12} \quad \text{subst in } (*)$$

$$\Rightarrow \tan \frac{\pi}{4} = 1 \quad \therefore (*) \Rightarrow (2)$$

$$1 - \tan \alpha \tan \beta - \tan \alpha \tan \gamma - \tan \beta \tan \gamma = \tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma$$

$$1 - \tan^2 \frac{\pi}{12} - \tan^2 \frac{\pi}{12} - \tan^2 \frac{\pi}{12} = 3 \tan \frac{\pi}{12} - \tan^3 \frac{\pi}{12}$$

$$\Rightarrow \tan^3 \frac{\pi}{12} - 3 \tan^2 \frac{\pi}{12} - 3 \tan \frac{\pi}{12} + 1 = 0$$

$\therefore \tan \frac{\pi}{12}$ is a root of

$$t^3 - 3t^2 - 3t + 1 = 0$$